

IRIS Studio: the cinema robot arm of the thesis as a browser simulator, with a servo zoom lens, a planned shot library and a learning loop

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Abstract

IRIS is the six-axis cinema robot arm of the author’s MEng thesis [1]: Unitree GO-M8010-6 actuators, an offset-axis layout, 0.7 m of reach and a visuomotor policy that learned camera moves from demonstrations. This document records the studio built on top of it in September 2026: the thesis’s MuJoCo kinematic model driven in the browser, a desk with product sets, a Sony α 7R III with a 16–150 mm zoom whose zoom and focus rings are servo-driven, a thin-lens depth-of-field model of the feed, 23 preset shots (pans, dollies, orbits, arcs, dolly zooms, rack focus, object tracking) defined as parametric lens paths, a damped-least-squares solver with a tool frame at the lens’s entrance pupil, and a demonstration-recording and policy-training loop. Every preset is planned at 30 Hz: the maximum joint speed over the library is $122^\circ/\text{s}$ and the peak inverse-dynamics torque 14.6 N m against the motors’ 23.7 N m, with 0 unreachable samples. The dolly zooms hold the subject’s image height to a coefficient of variation of 0.6 % and 0.5 % where a plain dolly changes it by 12.9 %. The videos on the web page were recorded from the same simulator; the executed telemetry (5739 frames over 23 shots) shows a mean lens position error of 0.26 mm and a mean aim offset of 11.29 % of the frame. All figures are regenerated by `experiments/make_figures.py`.

Keywords: cinema robotics, inverse kinematics, camera control, zoom lens, dolly zoom, imitation learning, browser simulation.

1 Overview

The thesis built IRIS, characterised it and learned camera moves from demonstration (VisionJoint-Planner: a five-frame window of RGB and joint states and a goal point in, the next five joint states out, trained on 17,897 samples). Its simulation lived in MuJoCo on a workstation and its camera was a RealSense. The studio described here moves the arm to the web page `iris.html` and its full-screen app `iris_studio.html`, adds the camera the next IRIS will carry, and turns cinematography into a library of planned, validated presets that a visitor can play, edit and learn from. The Python side (`experiments/irisstudio`) and the JavaScript side (`assets/js/iris`) implement the same kinematics and the same shot rules; the planner validates in Python what the studio executes in the browser, and the browser’s own execution is logged back for Section 3.4.

2 Method

2.1 Kinematics from the MuJoCo tree

`iris_model.json` is exported from `mujoco_sim/assets/iris.xml`: for every body its fixed pose in the parent (position, quaternion), its hinge joint (axis, range) and its mesh. MuJoCo composes $T_{\text{world},b} = T_{\text{world,parent}} T(\mathbf{p}_b, \mathbf{q}_b) R(\hat{\mathbf{a}}_b, q_j)$, and both the Python and the JavaScript forward kinematics do the same; `tests/test_kinematics.py` checks positions and rotations of four bodies against `mj_forward` to 10^{-9} over 50 random configurations, and the geometric Jacobian against `mj_jacBody`. Joint 1 at zero puts the arm’s plane along $+y$; -90° faces the products. Joints 4 ($\pm 180^\circ$) and 6 ($\pm 360^\circ$) are continuous-rotation motors: the solver leaves them unclamped and every path difference is taken modulo 2π (an early planner counted a $\pm 180^\circ$ wrap as a 358° jump).

The tool frame. The mount’s $+z$ is the optical axis and its $-y$ is up (the convention of the thesis’s look-at controllers). The camera body and lens are modelled to scale; the lens axis passes through the mount origin and the entrance pupil sits $\mathbf{o} = (0, 0, 0.14\text{m})$ ahead of it, so the tool frame is the mount frame translated by $R\mathbf{o}$. A first version aimed the mount origin and put the lens 2cm beside it, which placed the subject 10% off centre at close range; aiming the tool frame removed it. All shot positions are lens positions.

2.2 Inverse kinematics

Given a target lens position $\mathbf{p}^*$ and rotation R^* (a look-at rotation with a level horizon, or yaw/pitch/roll for pans and tilts), damped least squares iterates

$$\Delta \mathbf{q} = J^\top (JJ^\top + \lambda I)^{-1} \mathbf{e}, \quad \mathbf{e} = [\mathbf{p}^* - \mathbf{p}(\mathbf{q}); w \log(R^* R(\mathbf{q})^\top)^\vee], \quad \lambda = 10^{-3}, w = 0.3, \quad (1)$$

with a step of 0.6, at most 80 iterations from the previous joint vector, and a tolerance of 0.1mm and 0.06° . When the warm start fails (edge of reach), eight canonical postures are tried and the solution closest to the previous joint vector is kept; without a previous vector, the one whose base yaw faces the target with the largest margin to the joint limits. Unreachable poses make the studio hold its last pose and flag it, instead of clamping into a folded posture as an early build did. A per-joint speed cap of 1.5 rad/s (3 rad/s for the whip pan) filters the commanded joints; this is the “cinematic” rate limiter of the thesis’s deployment stack, not a dynamic model.

2.3 The lens

Full-frame sensor $35.9 \times 24\text{mm}$. Field of view $2 \arctan(24/2f)$ vertical. Depth of field from the thin-lens circle of confusion,

$$c(d) = \frac{f}{N} \frac{|d - S|}{d} \frac{f}{S - f}, \quad \text{acceptable } c = 0.03\text{mm}, \quad (2)$$

rendered in the feed by a Poisson-disk gather whose radius is c converted to pixels, with each tap weighted by its own circle so that sharp foregrounds do not bleed. Zoom and focus rings are servos slewing at 60 mm/s and 4 m/s with a short first-order lag; the barrel extends 35 mm over the zoom range. The dolly zoom commands $f(t) = f_0 d(t)/d_0$, which keeps a subject’s image height $h_{\text{img}} = f h/d$ constant while the arm moves the lens along the axis; the planner tracks h_{img} for every shot (Figure 5).

2.4 Shots

A preset is a path (static, line, horizontal orbit, vertical arc), an aim (look-at a subject or point, look-at moving between two points, yaw/pitch/roll ranges, or track the car), a lens track (focal length constant, ramped or dolly-zoom; focus automatic, fixed, ramped or a rack between subjects; aperture) and a duration with an easing (smoothstep, linear, or an exponential snap for the whip pan). Subjects are roles (*near* hero at $x = 0.32$ m, *far* at 0.74 m, extras, the car) resolved by the active product set, so every preset works on every set. The toy car drives a 0.18 m circle around (0.64, 0) in 12 s; the tracking preset aims at its true position, and the studio offers a pixel tracker (red-blob centroid on a 64×43 read-back of the feed, proportional gaze control) as the alternative.

2.5 The planner

`plan_shots.py` samples each preset at 30 Hz, solves the IK along it, unwraps the continuous joints, differentiates for joint speeds and accelerations, and calls `mj_inverse` on the thesis model (contacts disabled; the 1.5 kg camera payload is the model’s own `ee_mount` mass) for the joint torques. It reports per shot the IK residuals, the maximum joint speed and which joint, the peak torque and which joint, the closest approach to a joint limit, the lens ranges and the subject-size statistics, and exports the joint tracks for the page. Presets that failed the first drafts were re-sited with small parameter sweeps (Section 5).

2.6 Recording and learning

The studio records episodes at 10 Hz: joints, focal length, focus distance and a 24×16 grey thumbnail of the feed, tagged with the shot and the product positions. The in-tab policy is a VisionJointPlanner-lite in TensorFlow.js: five past states and the thumbnail through a two-layer CNN, a goal token (shot one-hot plus the hero and far positions), a 256-wide MLP with layer normalisation, and the next five joint and lens deltas (action chunking); at run time ACT-style temporal ensembling averages the chunks predicted at previous steps with weights $e^{-0.35j}$.

2.7 Video capture

`record_videos.js` drives the app in headless Chrome on the GPU through puppeteer, steps the simulation deterministically at 30 fps through `window.IRIS.stepFrame`, screenshots each frame of the film layout (studio view left, feed right, 1920×1080), encodes with `ffmpeg`, and stores the per-frame telemetry (joints, lens, IK status and residual, the aim point’s position in the frame) that Section 3.4 analyses.

3 Experiments

3.1 Lens model

3.2 Workspace and the shot library

Table 1 is the library as shipped: 23 presets, 178 s of motion, 0 unreachable samples, 2 presets above the comfort cap (the whip pan, by design), the largest speed $122^\circ/\text{s}$ and the largest torque 14.6 N m, at the shoulder as the thesis’s sizing predicts (25.8 N m worst case static with a 1.5 kg payload at full extension, which no preset reaches).

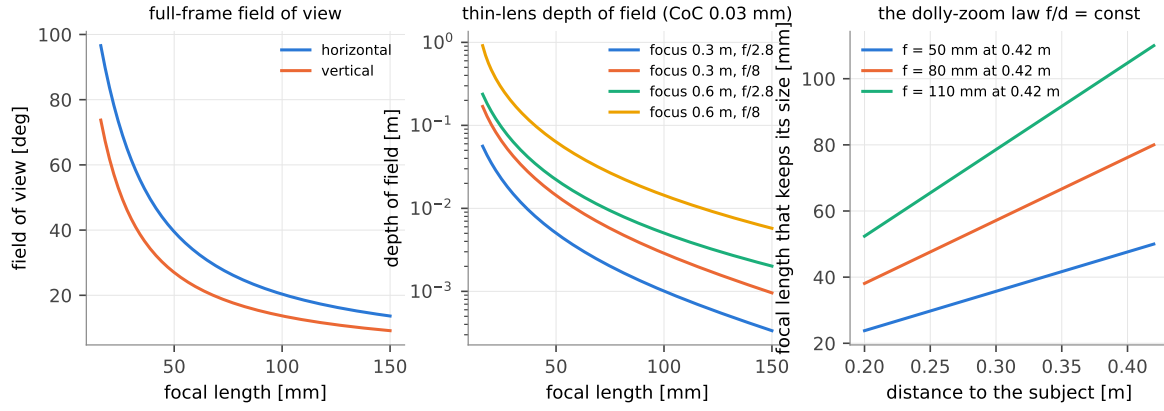


Figure 1: Field of view over the zoom range; depth of field for two focus distances and apertures; and the dolly-zoom law.

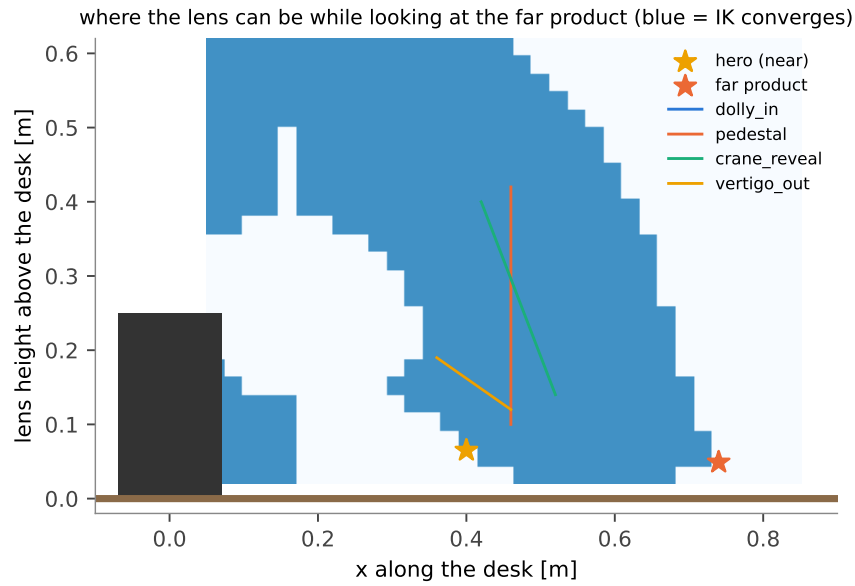


Figure 2: Lens positions in the vertical plane of the desk from which the far product can be looked at (IK converges), with four planned paths.

Table 1: Every preset through the planner (30 Hz): duration, focal length, maximum joint speed, peak inverse-dynamics torque, IK residuals, and the closest approach to a joint limit (continuous joints excluded).

shot	s	mm	$\dot{q}_{\max}$ [°/s]	$\tau_{\max}$ [N m]	pos [mm]	rot [°]	limit [°]
Pan	5	28	40	10.2	0.07	0.06	16
Tilt	5	28	29	7.9	0.10	0.06	15
Dutch roll	5	40	19	5.9	0.07	0.06	9
Zoom in	6	24–110	0	5.8	0.06	0.00	2
Zoom out	6	24–110	0	5.8	0.06	0.00	2
Dolly in	6	35	8	9.6	0.10	0.01	5
Dolly out	6	35	8	9.8	0.10	0.02	5
Truck (lateral)	6	35	100	14.6	0.10	0.02	25
Pedestal (crane)	6	35	42	11.0	0.10	0.02	25
Orbit 90°	8	35	65	5.6	0.10	0.05	15
Orbit 180°	14	35	75	10.3	0.10	0.05	11
Turntable 360°	14	45	0	5.9	0.06	0.01	29
Top view to side view	8	30	14	13.5	0.10	0.02	6
Crane reveal	7	28	31	10.3	0.10	0.03	37
Dolly zoom (out + zoom in)	8	70–98	6	9.8	0.10	0.02	5
Dolly zoom (in + zoom out)	8	68–95	6	9.6	0.10	0.01	5
Rack focus	8	55	0	3.9	0.10	0.01	9
Object tracking	12	45	46	6.1	0.10	0.06	10
Follow the car	12	40	55	6.4	0.10	0.06	5
Low reveal	6	24	63	8.7	0.10	0.06	19
Whip pan	4	35	122	13.2	0.09	0.06	13
Arc + zoom	12	30–60	83	6.4	0.10	0.05	2
Hero push-in	6	40–60	16	12.4	0.10	0.06	27

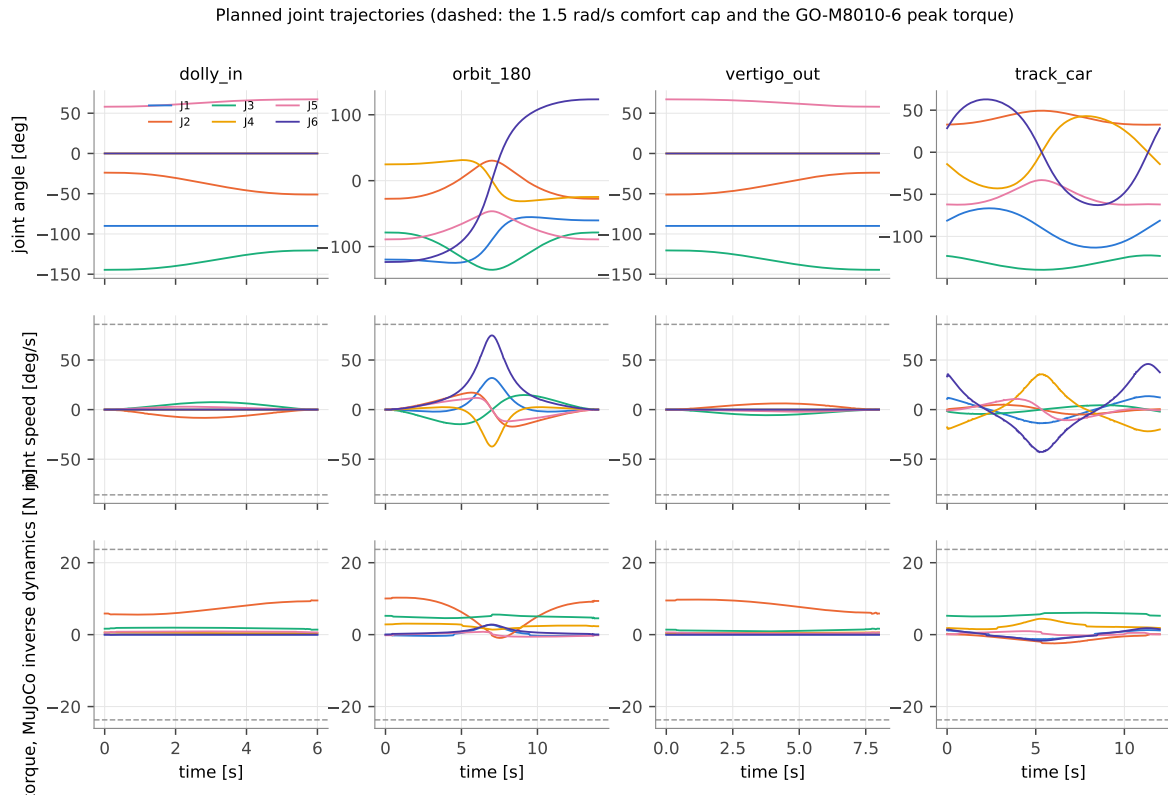


Figure 3: Four presets: joint angles, speeds against the 1.5 rad/s cap, and torques against the 23.7 N m peak.

Every preset through the planner: within the speed cap, the torque limit, and clear of the joint limits

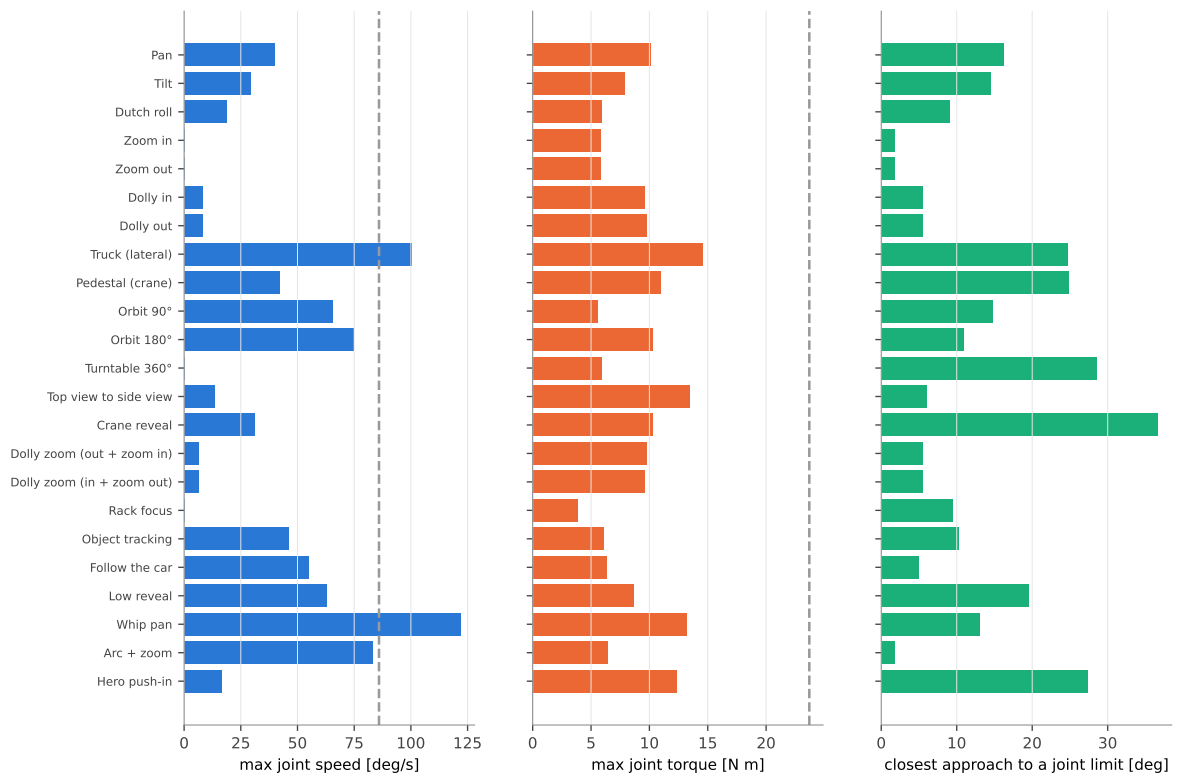


Figure 4: Per-shot maxima over the library.

3.3 Dolly zoom

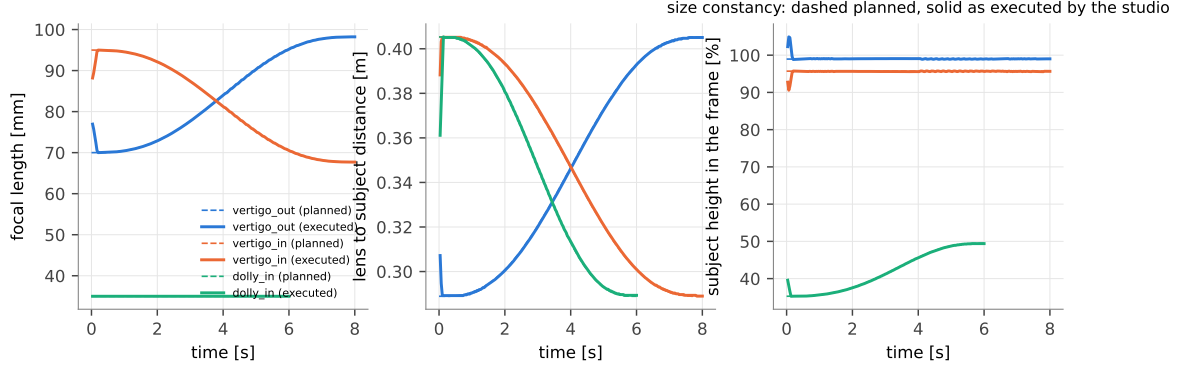


Figure 5: Focal length, lens distance and the subject’s image height along the two dolly zooms and a plain dolly.

The subject’s height in the frame varies by 0.6 % (out + zoom in) and 0.5 % (in + zoom out) along the dolly zooms, against 12.9 % for the plain dolly over the same distance; the residual comes from the servo lag and the discrete focus of the lens model.

3.4 Executed telemetry

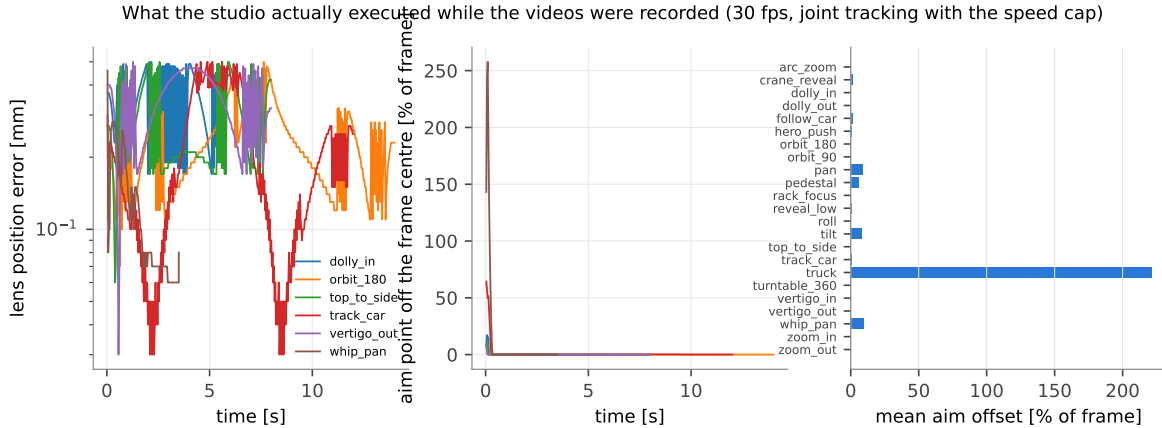


Figure 6: From the video recordings: the live solver’s lens position error and the aim point’s offset from the frame centre.

Over 5739 recorded frames of 23 shots the live solver reports a mean position error of 0.26 mm (maximum 0.5 mm at the fast whip) and 0 frames out of reach; the aim point sits 11.29 % of the frame from the centre on average (maximum 17497.2 %), the difference between the commanded look-at and the rate-limited joints.

4 Implementation and usage

experiments/irisstudio/kinematics.py (Arm, tool frame, IK, seeds), shots.py (the evaluator), plan.shots.py, record_videos.js, make_figures.py; the spec out/iris_studio.json is copied

to the site as `assets/data/iris_studio.json`. Browser: `assets/js/iris/kin.js`, `lens.js`, `shots.js`, `scene.js` (three.js: desk, arm from the decimated STL links, camera rig, products, lights), `dof.js`, `policy.js`, `app.js`; pages `iris_studio.html` and `iris.html`.

```
cd experiments && python -m pytest -q tests
python plan_shots.py                # out/shots_plan.json, assets/data/iris_shots_plan.json
node record_videos.js               # out/video/iris_<shot>.mp4 + telemetry (needs http.server)
python make_figures.py              # figures, numbers.tex, shots_table.tex
cd ../docs && make
```

Hardware path: the planner’s joint tracks are 30 Hz position setpoints in the model’s joint order; the thesis’s ROS node consumes exactly that over RS-485, and the lens tracks are two servo setpoints per frame.

5 Lessons and limitations

- Orbits at desk height fail: looking back toward the base needs the wrist to fold past its $\pm 100^\circ$ pitch. The turntable orbit lives high above the product and looks down; half orbits stay on the base side.
- A straight-down top view sits on the look-at singularity (the horizon has no meaning); the top-to-side arc starts at 70° elevation and comes in from the side.
- Any lens position closer than about 0.3 m to the base column folds the arm onto itself once the tool offset is included; the first drafts of the dollies started there.
- The wrist singularity ($q_5 \rightarrow 0$) makes joints 4 and 6 swing quickly when the optical axis passes through the forearm axis; the tracking presets were re-sited to keep $|q_5|$ away from zero, and continuous joints are unwrapped.
- Not modelled: motor dynamics and belt compliance, self-collision, lens breathing and the real minimum-focus curve, exposure. The in-tab policy is a toy; the thesis’s numbers stand.

References

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